Alpha Spectroscopy

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Abstract

In this experiment, a radioactive ^{226}Ra source was used to study the decay chain of the source using silicon detectors. Using the source the energy calibration was done using first four peaks and fifth peak was used to verify the validity of the calibration. The measured activity of the source is 1.376(22) kBq and the calculated age of the source is 17.51(201) years. The analysis of counting statistics does give measured root mean square value well within 2 σ, where σ is $\sqrt{N}$.

1 Background Introduction

1.1 Motivation

A decay chain refers to the sequence in which a parent radioactive nucleus decays into stable daughter nuclei through a series of unstable intermediate stages. The time required for the parent nucleus to fully decay into a stable form can vary widely, enabling scientists to estimate its age. The decay of identical nuclei over time follows the exponential decay law $e^{-\lambda t}$, where λ is a decay constant. The decay constant can be used to calculate the half-life and activity of the source and eventually the age of the parent nuclei. There are mainly four types of decay, alpha decay, beta decay, inverse beta decay, and isomeric transition. This experiment focuses on the alpha decay of the radium source.

1.2 Summary of experiments

This experiment was divided into three parts. The first part was to find the calibration of the alpha energy used in the experiment. After that, the next goal was to confirm the statistical precision of measurement of N counts is $\sqrt{N}$. For that, many data runs were taken to get root mean square of the count of each peak. And the root mean square of each peak was compared to $\sqrt{N}$ to test whether statistical precision of measurement of N counts is $\sqrt{N}$. And, finally the activity and age of the source were calculated and compared to the value given in the lab manual.

2 Theory

The decay of a particle follows the exponential decay function and the number of particle at time 't' can be calculated using:

$$N_A = N_A(0)\, e^{-t/\tau_A} \tag{1}$$

Where $N_A(0)$ is the number of the parent particle at $t = 0$ and $\tau_A = \frac{t_{1/2}}{\ln(2)}$. And, the activity is defined as the number of decay per unit time and is given by:

$$-\frac{dN_A}{dt} = \frac{N_A}{\tau_A} \tag{2}$$

When a particle decays into daughter nuclei, the daughter nuclei also undergoes decaying with its own half-life. The total number of daughter nuclei can be calculated by using the following equation:

$$N_B = C\, e^{-t/\tau_A} + D\, e^{-t/\tau_B} \tag{3}$$

In this experiment, we are starting with pure ^{226}Ra source, the above equation can be written as:

$$N_B = \frac{N_A(0)}{\tau_A\ \tau_B - 1}\,(e^{-t/\tau_A} - e^{-t/\tau_B}) \tag{4}$$

Using Equation 2 and 4, one gets simple Bateman equation:

$$R = \frac{N_B\,\tau_A}{N_A\,\tau_B} = \frac{\tau_A}{\tau_A - \tau_B}\,(1 - e^{-t/\tau_B + t/\tau_A}) \tag{5}$$

To find the age of the source, the above equation is inverted to get:

$$t = -\frac{\tau_A\,\tau_B}{\tau_A - \tau_B}\,\ln\left(1 - R\frac{\tau_A - \tau_B}{\tau_B}\right) \tag{6}$$

For analysis of this experiment, $\frac{dN_A}{dt}$ is calculated first by counting the total number of counts in the first peak then that number is divided by total time it was used for the data runs. This number was the activity of the source without considering the efficiency of the detector. Multiplying this resulted number by τ_A, N_A was calculated. Similar procedure was followed for the calculation of N_B. Once N_A and N_B were calculated, R and t can be calculated using Equation 5 and 6 respectively.

To calculate the activity of the source by first the efficiency of the detector was calculated. The efficiency of the detector is given by:

$$\eta = \frac{S}{4\pi R^2} \tag{7}$$

Using Equation 2 and 7, the activity of the source was calculated. And, it is given by:

$$A_{\text{source}} = \frac{N_A}{\tau_A\,\eta} \tag{8}$$

3 Experimental Methods

3.1 Description of Apparatus

The experimental setup for this experiment is shown in Figure 1. The radium source was placed in the center of an evacuated chamber 89.5 mm away from the silicon detector, and the high-voltage detector was tuned to 80 V. The signal from the silicon detector was passed into an amplifier where the output was sent to a timing scaler which we used to strobe the PiDAQ. A total of 18 runs were performed: 1 calibration run, 1 background run, and 16 data runs. The amplifier settings used in the lab are given in Table 1.

Coarse Gain	Fine Gain
200	5.0

Table 1: Amplifier settings.

The amplifier settings used in the lab are given in Table 2

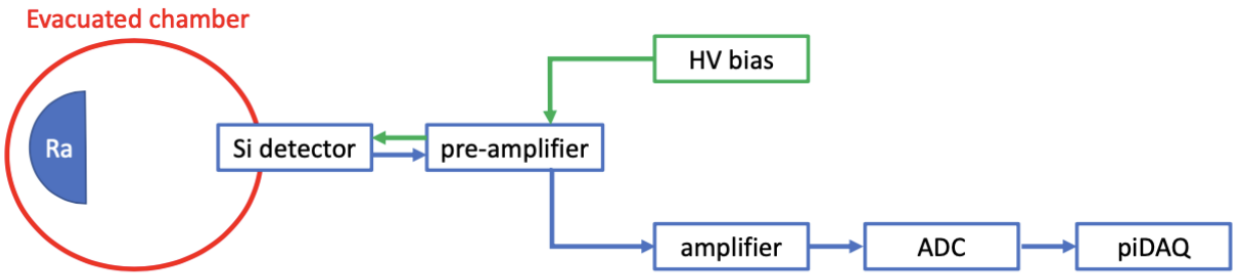


Figure 1: Experimental setup for α-Spectroscopy experiment.

Upper Level	Lower Level
0	0.5

Table 2: ORTEC 522 Settings.

3.2 Description of experimental procedure

Once the radium source was placed in the evacuated chamber, the calibration data was taken for 10 minutes. The background run was taken for 5 minutes to see the general shape of the background. Once the calibration data and background data were taken, the rest of the experiment was performed. For the rest of the experiment, each run was for 600 seconds.

3.3 Description of the observed phenomena

Once the data were taken, five peaks were observed in the alpha-energy spectra. The first peak is from the 4.7 MeV alpha energy, the second is the 5.3 MeV alpha energy, the third is the 5.4 MeV alpha energy, the fourth is the 6.0 MeV alpha energy, and the fifth is the 7.6 MeV alpha energy. This alpha-energy data was only analyzed for calibration of the peaks.

The first peak is due to the α-decay of ^{226}Ra to ^{222}Rn, while the second peak is due to the α -decay of ^{210}Po to ^{206}Pb. In the decay of ^{222} Rn to ^{218} Po, the third peak is obtained, while the decay of ^{218}Po to ^{214}Pb results in the fourth peak. Finally, the fifth peak is the result of the decay of α of ^{214} Po to ^{210} Pb.

4 Analysis and Results

4.1 Description of the data set

The data were recorded on the computer using the PiDAQ system. Then they were converted to the xy format for further analysis. The tag number was given to the data to correctly identify them for analysis. The tag0005 has data for the calibration run, while tag0006 has data for the background run. The tag0007-tag0022 are used for the data runs.

4.2 Method of Analysis

The analysis was done using Python. The calibration of the alpha energy was performed using first four peaks, and then using this calibration it was checked whether the calibration correctly gives the energy of the fifth peak. The calibration was done by first finding the peak position of the peak and then use 'curve fit' function from the Scipy.

Once the calibration was done, the statistical precision on a measurement of counts for each peak was done. The gaussian spectrum of each peak for for all the tags from tag0007 to tag0022 was plotted. It was found that the gaussian distribution of each peaks starts to shift significantly after tag0017 which can be seen in the Figure 2. So, for the rest of the analysis only data from tag0007 to tag0017 are used.

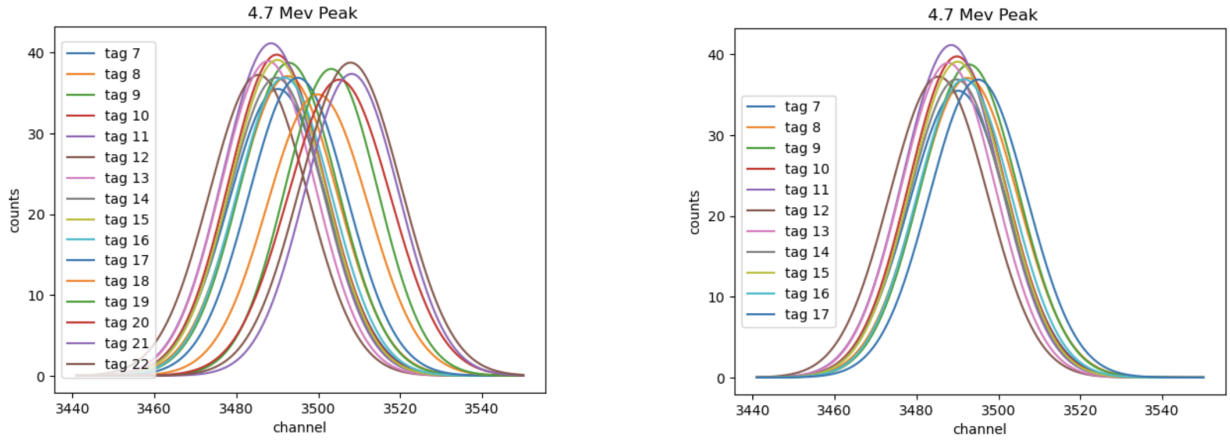


Figure 2: 4.7 MeV peak for different data runs.

So, for each peak the counts was counted by summing all the counts for that peak from tag0007 to tag0017. The background for that peak was also subtracted using background run. Since there are 11 tags that are being used for the analysis. The background is subtracted 11 times: one background subtraction for each tag. For each of these peaks, the rms on the counts of 11 runs was calculated and then the square root of the total count was also calculated to check the counting statistics for the data. Once this was done for each peak, the same calculation was done for the whole spectrum.

Once the counting statistic analysis was done, the source activity and its age was calculated using just first two peaks. The first peak helps in finding the activity of the source, while the first two peaks help in finding the age of the source. Once the source activity and its age was calculated, it was compared to their actual value given in the lab manual.

4.3 Presentation of results

Calibration: The calibration of alpha energy is shown in the Figure 3:

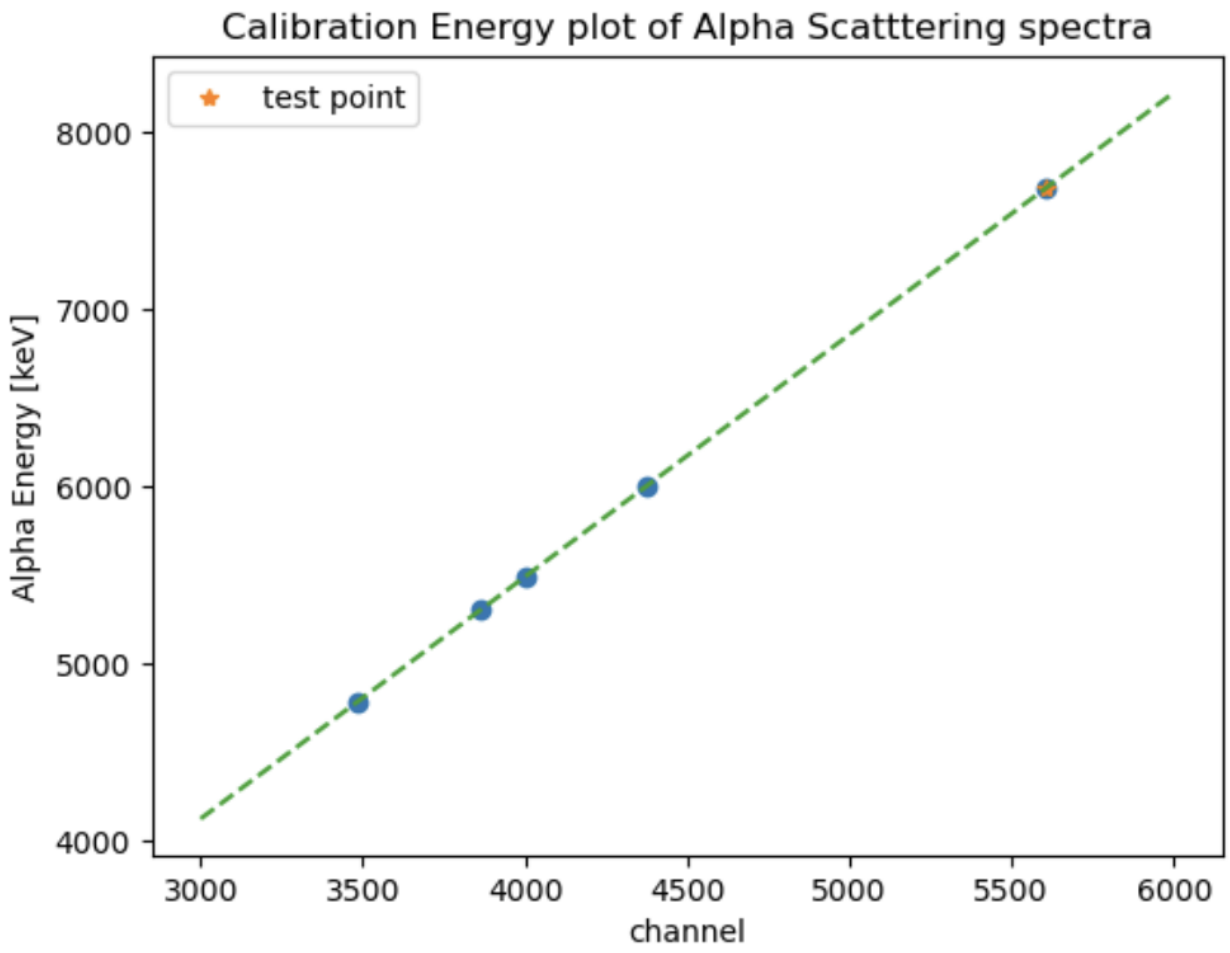


Figure 3: Calibration of α energy for 5 peaks.

The channels and peak position used for calibration are given by Table 3

Table 3: Energy calibration peak positions.

	4.7 MeV Peak	5.3 MeV Peak	5.4 MeV Peak	6.0 MeV Peak	7.6 MeV Peak
channel	3483.15 ± 0.37	3862.22 ± 0.70	3999.24 ± 0.43	4373.85 ± 0.43	5605.24 ± 0.49
Calibration Energy (keV)	4784.34 ± 0.25	5304.33 ± 0.07	5489.48 ± 0.30	6002.55 ± 0.10	7686.82 ± 0.10

The fit parameters from fitting this data from Table 3 is given in the Table 4

Statistical Precision Measurement

The result for statistical precision measurement is presented in table 5

Activity and age of the source : The activity and age of ^{226}Ra source is given in the Table 6

4.4 Discussion of results

4.4.1 Calibration

The calibration was done using first four peaks and using the fit-parameter the fifth peak was calculated to check whether the calibration is reasonable or not. From Figure 3, it can be seen that fit function does give the correct alpha energy for the last peak.

4.4.2 Statistical Precision Measurement

From Table 5, it can be seen that 5.4 MeV peak, 6.0 MeV peak, and the whole spectrum do agree that the precision of N counts is $\sqrt{N}$. The other peak also fall within 2 σ of N counts, where σ is $\sqrt{N}$.
As we could only use 10 sets of data for this analysis, the results are not as expected. The result of counting statistics gets better if you have more sample space. For this experiment, it requires more number of data sets to see all the peaks following the principle of counting statistics.‘

4.4.3 Activity and age of the source

From Table 6, it can be seen that there is large difference between measured activity and expected activity. This could be due to various underlying systematic error that we could not incorporate in our calculations. But it does give the expected age of the source within 1 σ of the measured age as can be seen in the Table.

Table 4: Energy calibration fit parameters.

Slope (keV/Channel)	Intercept (keV)
1.366 ± 0.002	29.36 ± 6.51

Table 5: Statistical Precision Measurement .

	counts	Measured RMS	Expected $\sqrt{N}$	% difference
4.7 peak	13579	44.25	35.13	-25.93
5.3 peak	5868	26.27	24.22	-8.43
5.4 peak	12946	26.61	35.98	26.03
6.0 peak	13293	26.94	36.46	26.11
7.6 peak	12596	37.09	35.49	-4.49
Whole Spectrum	61762	52.77	74.93	29.57

Table 6: Activity and age of ^{226}Ra source.

η	0.00298
Measured Activity	1.376 ± 0.022 kBq
Expected Activity	3.24 kBq
Measured Age	17.51 ± 2.01 years
Expected Age	18.10 years
Difference	0.58 years

5 Conclusions

In this experiment, ^{226}Ra decay chain was explored to find the activity and age of the source. The age of the source was found to be 17.51 years which is well within 1 σ of the expected age of the ^{226}Ra source.

This decay chain was also used to analyze the precision of counting statistics. Though most peaks agree with the principle of counting statistic, 4.7 MeV peak only agrees to it within 26%. This result could have been improved using more sets of data run.

References

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